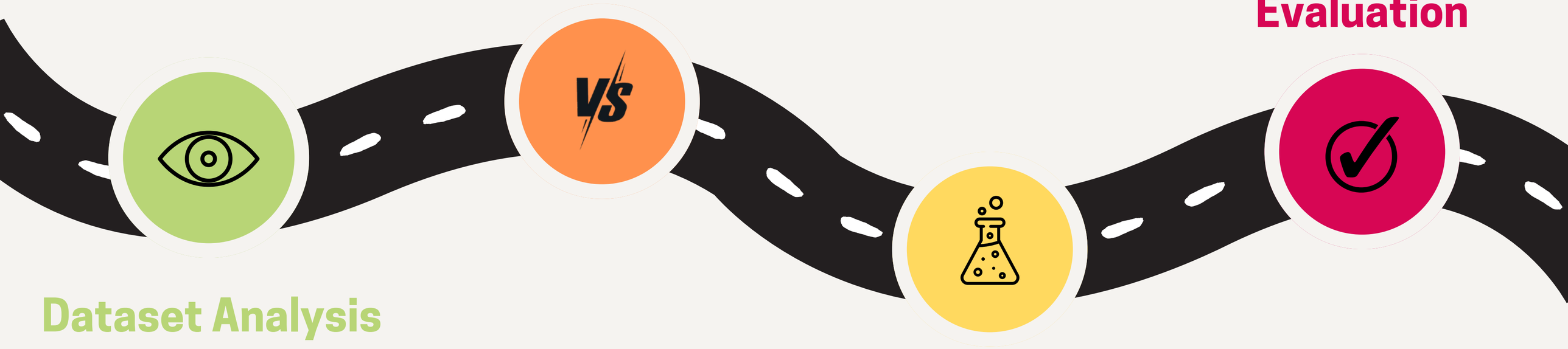


WASTE IDENTIFICATION

PROJECT'S ROADMAP

Model Comparison

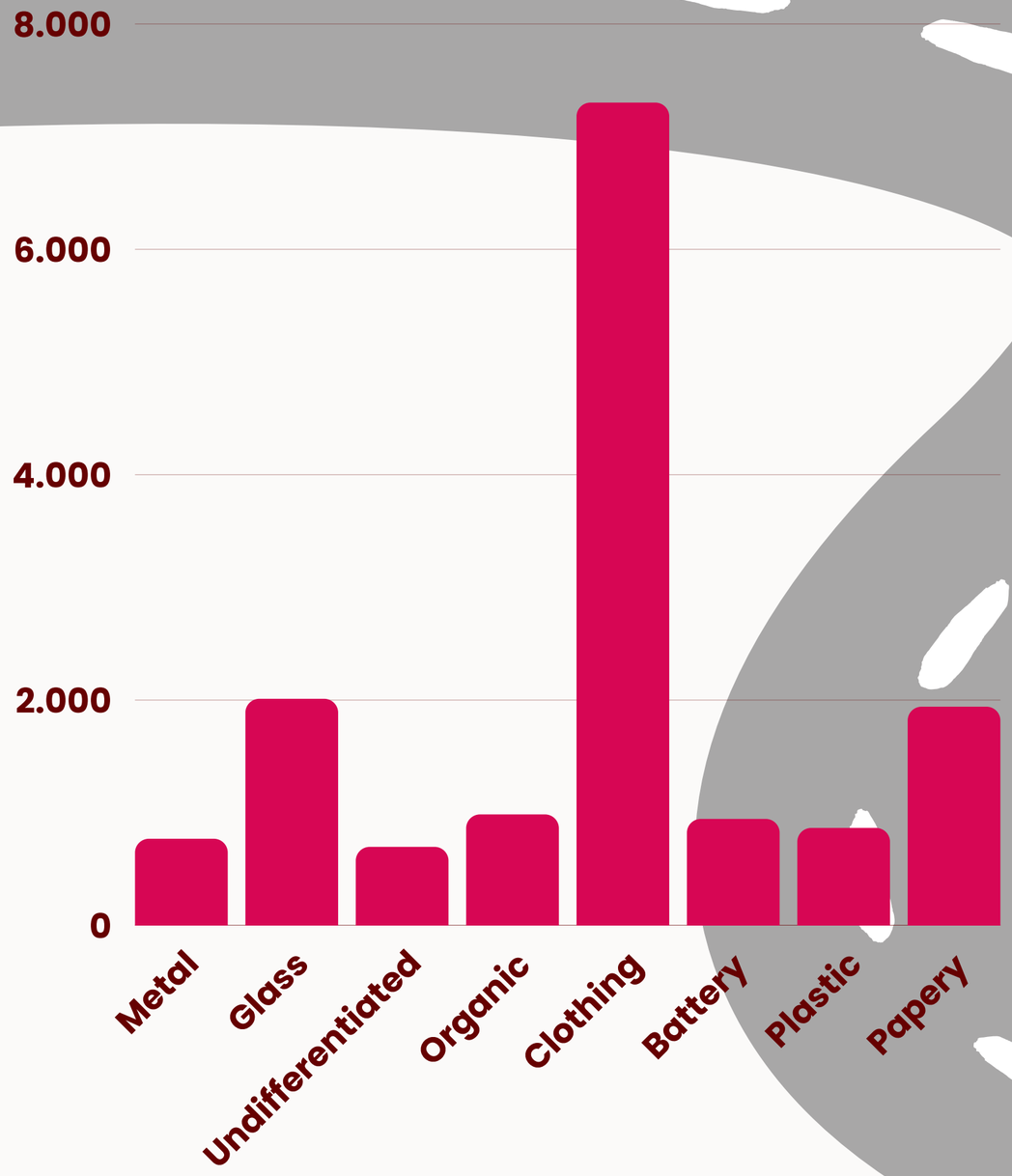
Results & Outcome Evaluation



Dataset Analysis

Experimental Optimization

DATASET ANALYSIS



The Problems

Severe Imbalance
Accuracy Paradox
Generalization Drop

The Solutions

Targeted Data Augmentation
Inverse Scaling
Weighted Loss

MODEL COMPARISON

Trash-Net

Transformer Architecture

Transformers require massive datasets to prevent overfitting, exceeding the project's data capacity

MODEL COMPARISON

YOLOv5

Object Detection Model

The project's objective is pure Image Classification; YOLOv5 would introduce unnecessary and expensive layers.

MODEL COMPARISON

ResNet-18

Baseline Model

Selected as the project baseline and customized with a Weighted Cross-Entropy loss

MODEL COMPARISON

EfficientNet-B0

Top Performer

Selected for advanced optimization and secondary analysis phase to improve the baseline

Softmax

Residual Mapping

Skip Connections

Gradient Stability

Transfer-Learning

ResNet-18

Architecture

Learning-Rate

Dropout-Rate

Weight-Decay

Layers

Loss Type

LR Scheduler

Augmentation

ResNet-18

Hyperparmeters

Learning Rate
0.0001

Batch Size
32

Loss Type
Weighted Cross Entropy

Agumentation
Uniform

LR Scheduler
ReduceLROnPlateu

Scheduler Patience
2

Layers Unlocked
1

Scheduler Factor
0.1

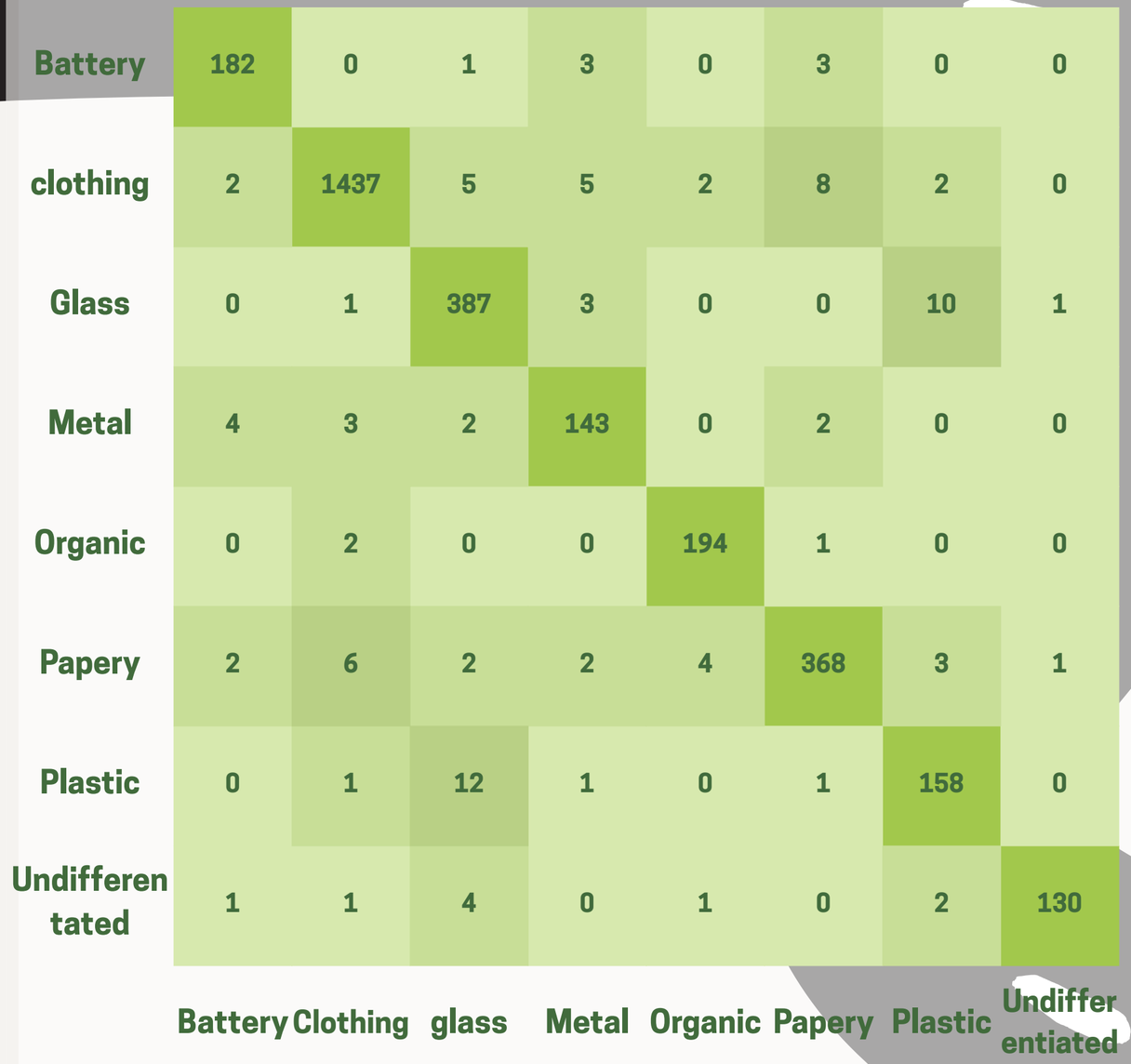
Early Stopping
5 epochs

Optimizator
Adam

ResNet-18

Baseline

BASELINE ANALYSIS



	Battery	Clothing	glass	Metal	Organic	Papery	Plastic	Undifferentiated
Battery	182	0	1	3	0	3	0	0
clothing	2	1437	5	5	2	8	2	0
Glass	0	1	387	3	0	0	10	1
Metal	4	3	2	143	0	2	0	0
Organic	0	2	0	0	194	1	0	0
Papery	2	6	2	2	4	368	3	1
Plastic	0	1	12	1	0	1	158	0
Undifferentated	1	1	4	0	1	0	2	130

Validation Loss

0,1589

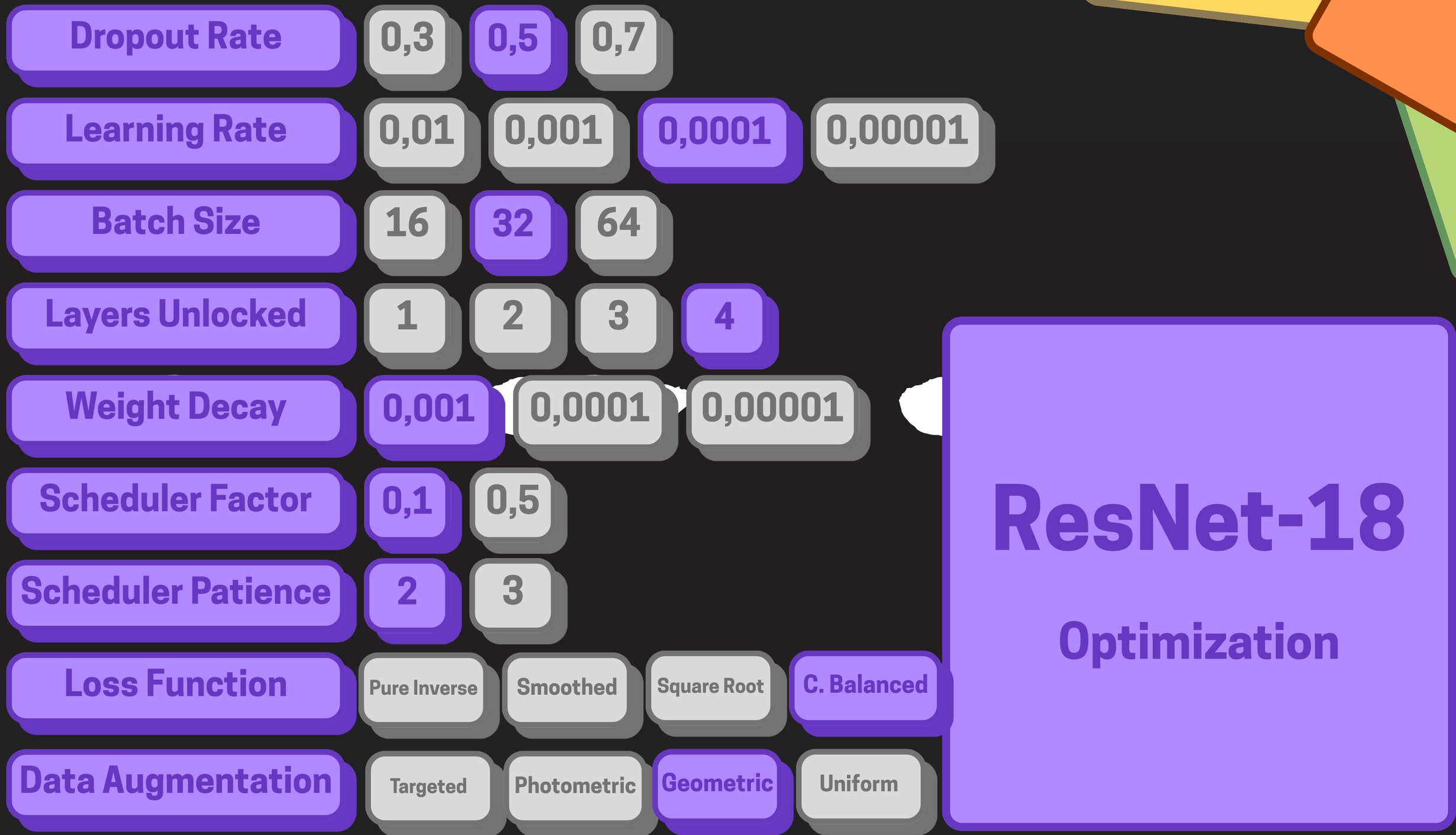
Balanced Accuracy

0,9524

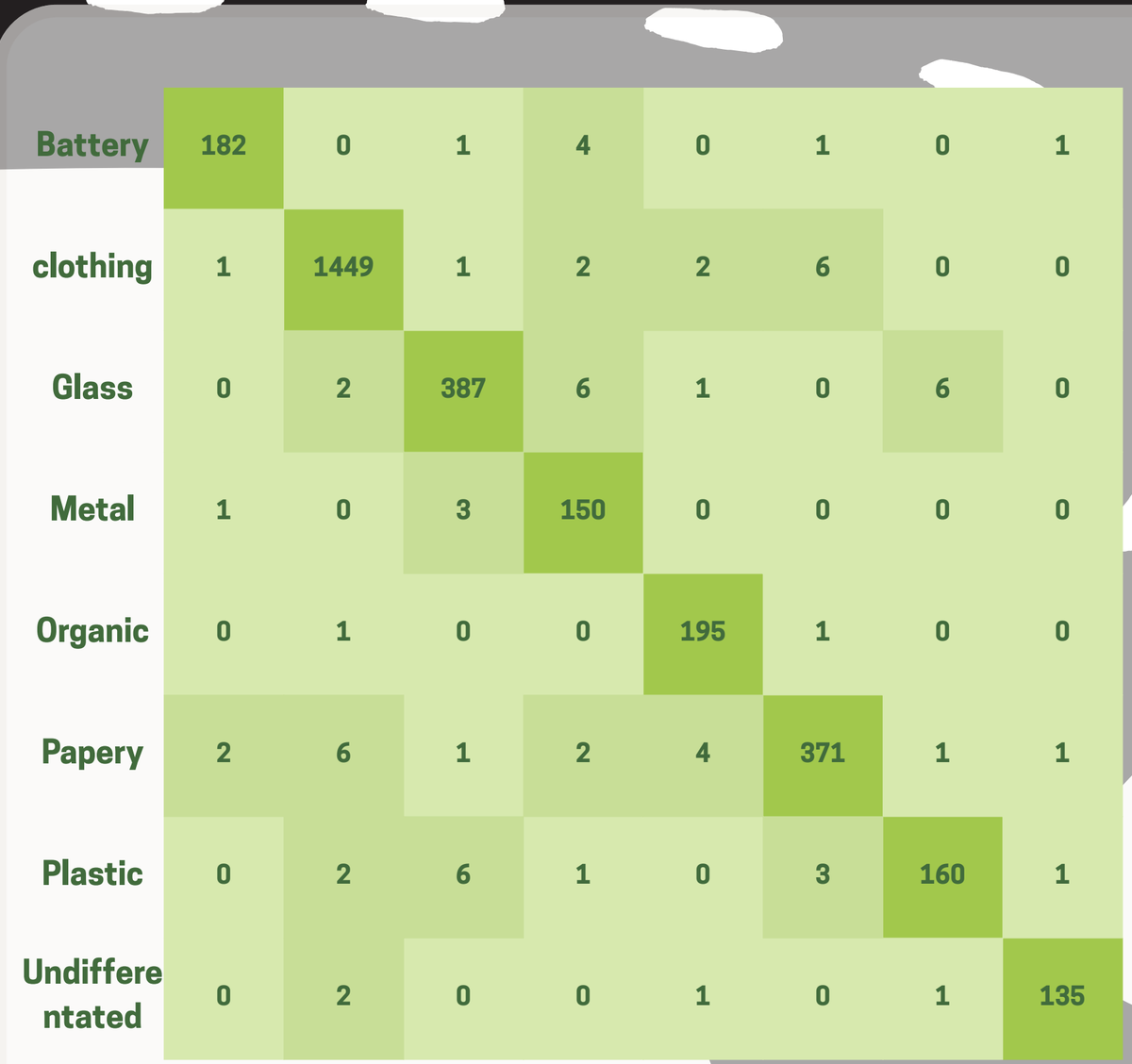
F1-Score

0,9514


Dropout Rate
0,3
0,5
0,7
Learning Rate
0,01
0,001
0,0001
0,00001
Batch Size
16
32
64
Layers Unlocked
1
2
3
4
Weight Decay
0,001
0,0001
0,00001
Scheduler Factor
0,1
0,5
Scheduler Patience
2
3
Loss Function
Pure Inverse
Smoothed
Square Root
C. Balanced
Data Augmentation
Targeted
Photometric
Geometric
Uniform
ResNet-18
Optimization


Dropout Rate
0,3
0,5
0,7
Learning Rate
0,01
0,001
0,0001
0,00001
Batch Size
16
32
64
Layers Unlocked
1
2
3
4
Weight Decay
0,001
0,0001
0,00001
Scheduler Factor
0,1
0,5
Scheduler Patience
2
3
Loss Function
Pure Inverse
Smoothed
Square Root
C. Balanced
Data Augmentation
Targeted
Photometric
Geometric
Uniform
ResNet-18
Optimization

OPTIMIZATION ANALYSIS



Validation Loss

0,0772

Balanced Accuracy

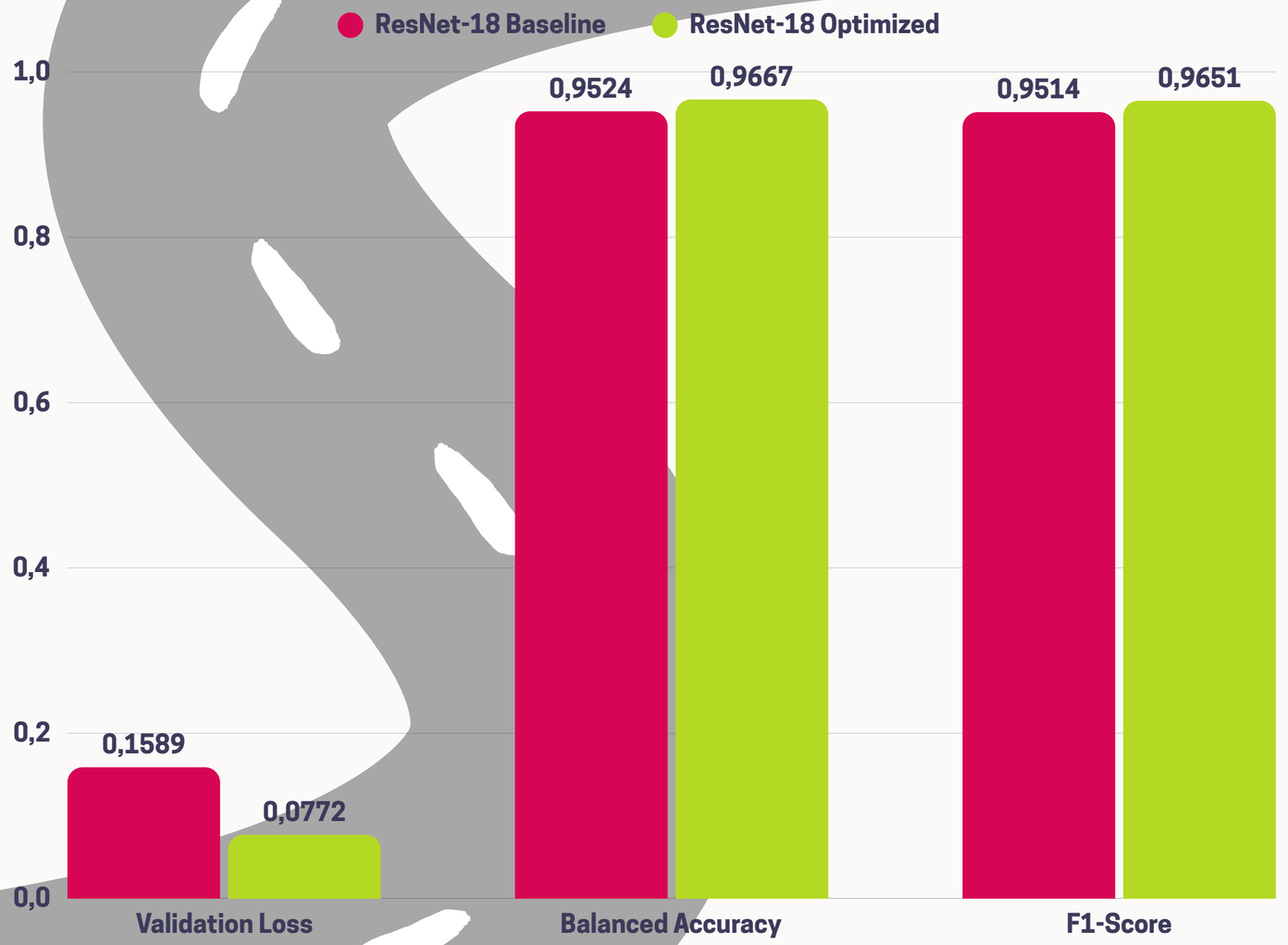
0,9667

F1-Score

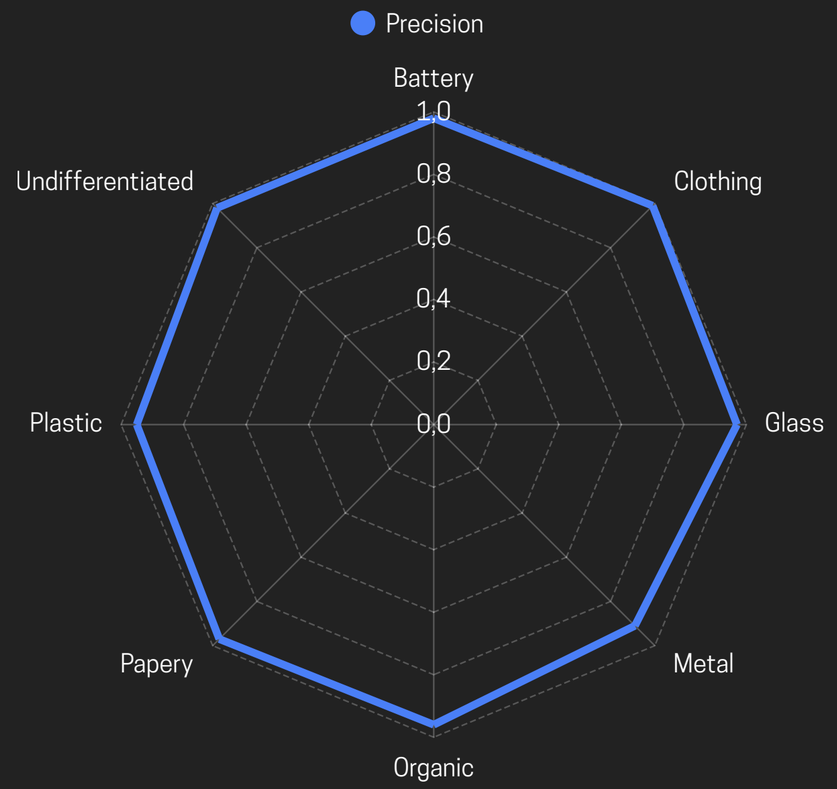
0,9651

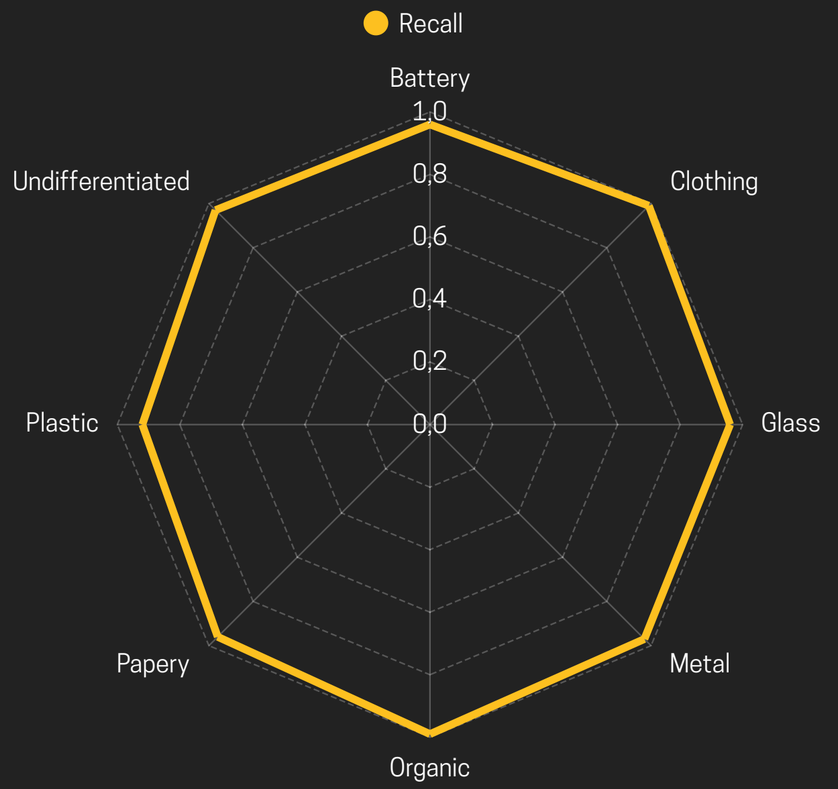
	Battery	clothing	Glass	Metal	Organic	Papery	Plastic	Undifferentated
Battery	182	0	1	3	0	3	0	0
clothing	2	1437	5	5	2	8	2	0
Glass	0	1	387	3	0	0	10	1
Metal	4	3	2	143	0	2	0	0
Organic	0	2	0	0	194	1	0	0
Papery	2	6	2	2	4	368	3	1
Plastic	0	1	12	1	0	1	158	0
Undifferentated	1	1	4	0	1	0	2	130

	Battery	clothing	Glass	Metal	Organic	Papery	Plastic	Undifferentated
Battery	182	0	1	4	0	1	0	1
clothing	1	1449	1	2	2	6	0	0
Glass	0	2	387	6	1	0	6	0
Metal	1	0	3	150	0	0	0	0
Organic	0	1	0	0	195	1	0	0
Papery	2	6	1	2	4	371	1	1
Plastic	0	2	6	1	0	3	160	1
Undifferentated	0	2	0	0	1	0	1	135



OPTIMIZED RESNET'S DETAILS







RESNET'S LIMITS



Rigid Structure

Difficulty in extracting complex micro-textures for similar materials



Dying ReLU

The ReLU function creates sharp angles at zero, risking the "dying neuron" phenomenon for critical nodes during training



High Parameter Count

11 Million parameters for a performance level that left margins for further optimization

THE TRANSTION

ResNet-18

- Architecture: Standard Residual Blocks.
- Activation: ReLU (Sharp angle at zero).
- Efficiency: 11M+ Parameters.

Fixed structure that limits adaptability to more complex materials.

EfficientNet-B0

- Architecture: MBConv (Inverted Bottleneck + Squeeze-and-Excitation).
- Activation: SiLU/Swish.
- Efficiency: ~5.3M Parameters (50% Reduction).

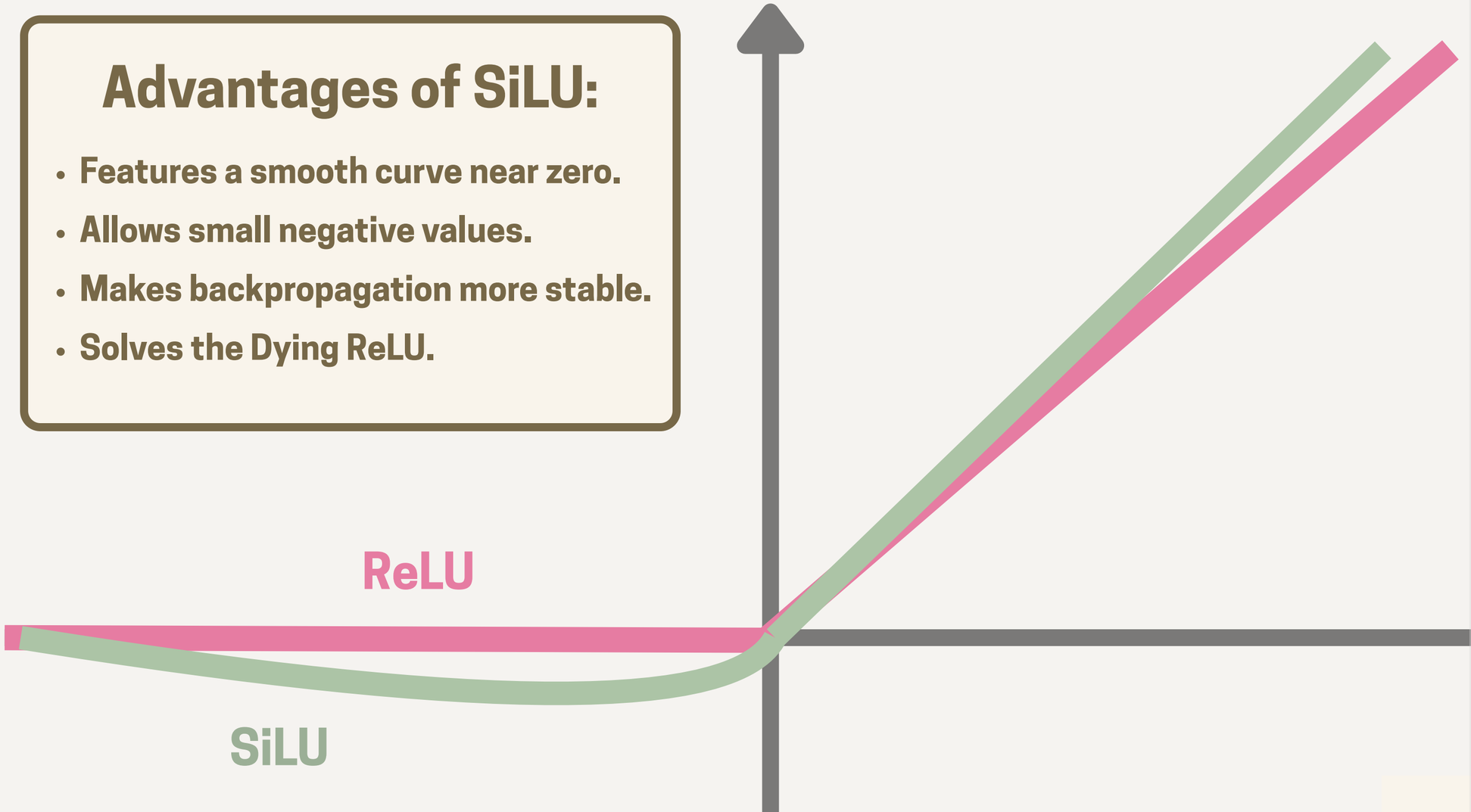
Superior feature extraction using half the computational resources."

EFFICIENTNET

From ReLU to SiLU

Advantages of SiLU:

- Features a smooth curve near zero.
- Allows small negative values.
- Makes backpropagation more stable.
- Solves the Dying ReLU.

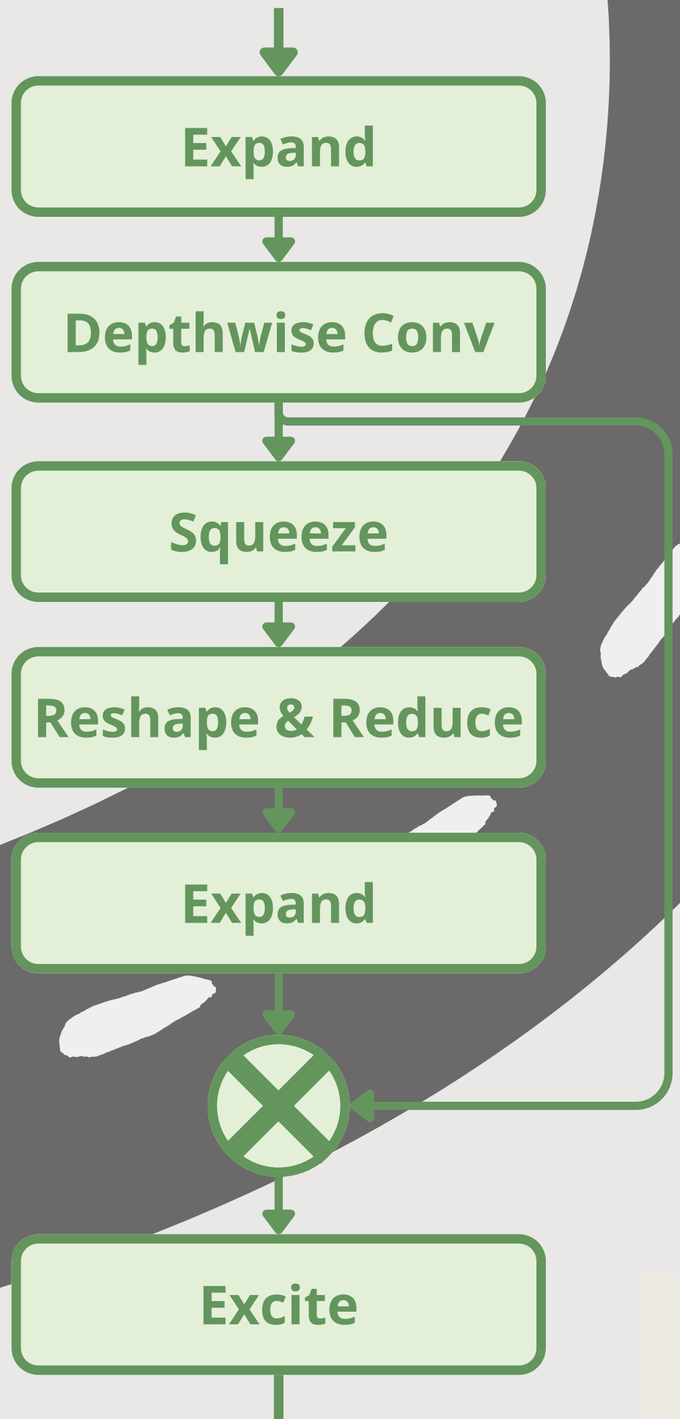


EFFICIENTNET

MBConv

The Mobile Inverted Bottleneck Convolution block redefines efficiency:

- **Projects channels into a higher-dimensional space.**
- **Lightweight yet powerful feature extraction.**
- **Recalibrates channel-wise feature responses (Squeeze-and-Excitation)**
- **Reduces dimensionality without losing critical information.**



SiLU / Swish

MBConv Block

Inverted Bottleneck

Depthwise Convolution

Linear Bottleneck

Squeeze-and-Excitation

EfficientNet-B0

Architecture

EfficientNet-B0

Advantages

- 5.3M Parameters
- -50% Parameters
- Low Computational Cost
- GPU Memory Optimization
- Hardware-Constrained

Batch Size

Dropout-Rate

Learning-Rate

Weight-Decay

Layers

WCE

Label Smoothing

Scheduler Patience

Augmentation

EfficientNet-B0

Hyperparameters

Learning Rate
0.0001

Batch Size
32

Loss Type
Class-Balanced Loss

Dropout Rate
0.5

LR Scheduler
ReduceONPlateu

Agumentation
Geometric Only

Layers Unlocked
All

Scheduler Patience
2

Weight Decay
0,001

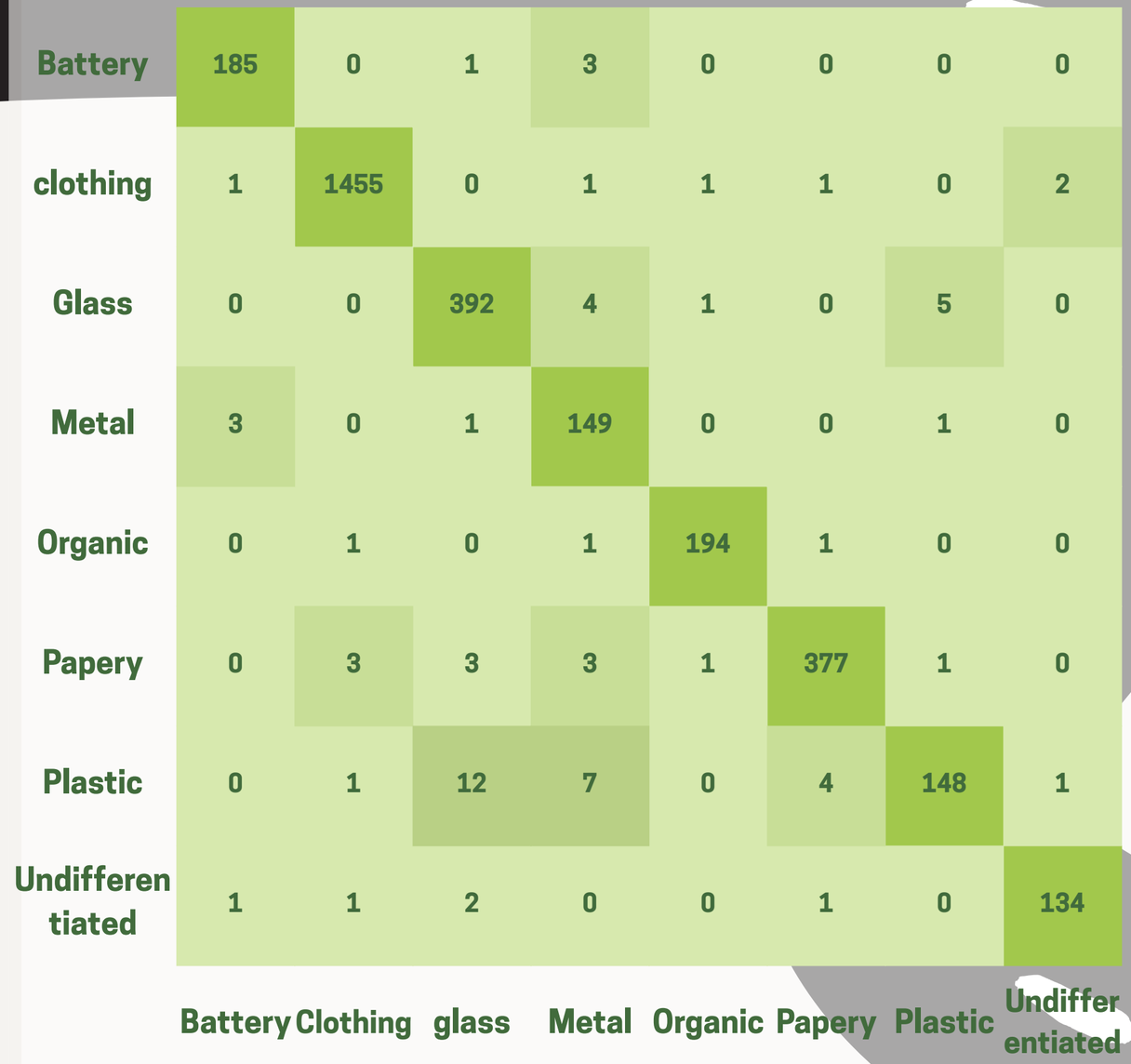
Scheduler Factor
0.1

EfficientNet-B0

Baseline

Direct Hyperparameter Transfer

BASELINE ANALYSIS



	Battery	Clothing	glass	Metal	Organic	Papery	Plastic	Undifferentiated
Battery	185	0	1	3	0	0	0	0
clothing	1	1455	0	1	1	1	0	2
Glass	0	0	392	4	1	0	5	0
Metal	3	0	1	149	0	0	1	0
Organic	0	1	0	1	194	1	0	0
Papery	0	3	3	3	1	377	1	0
Plastic	0	1	12	7	0	4	148	1
Undifferentiated	1	1	2	0	0	1	0	134

Validation Loss

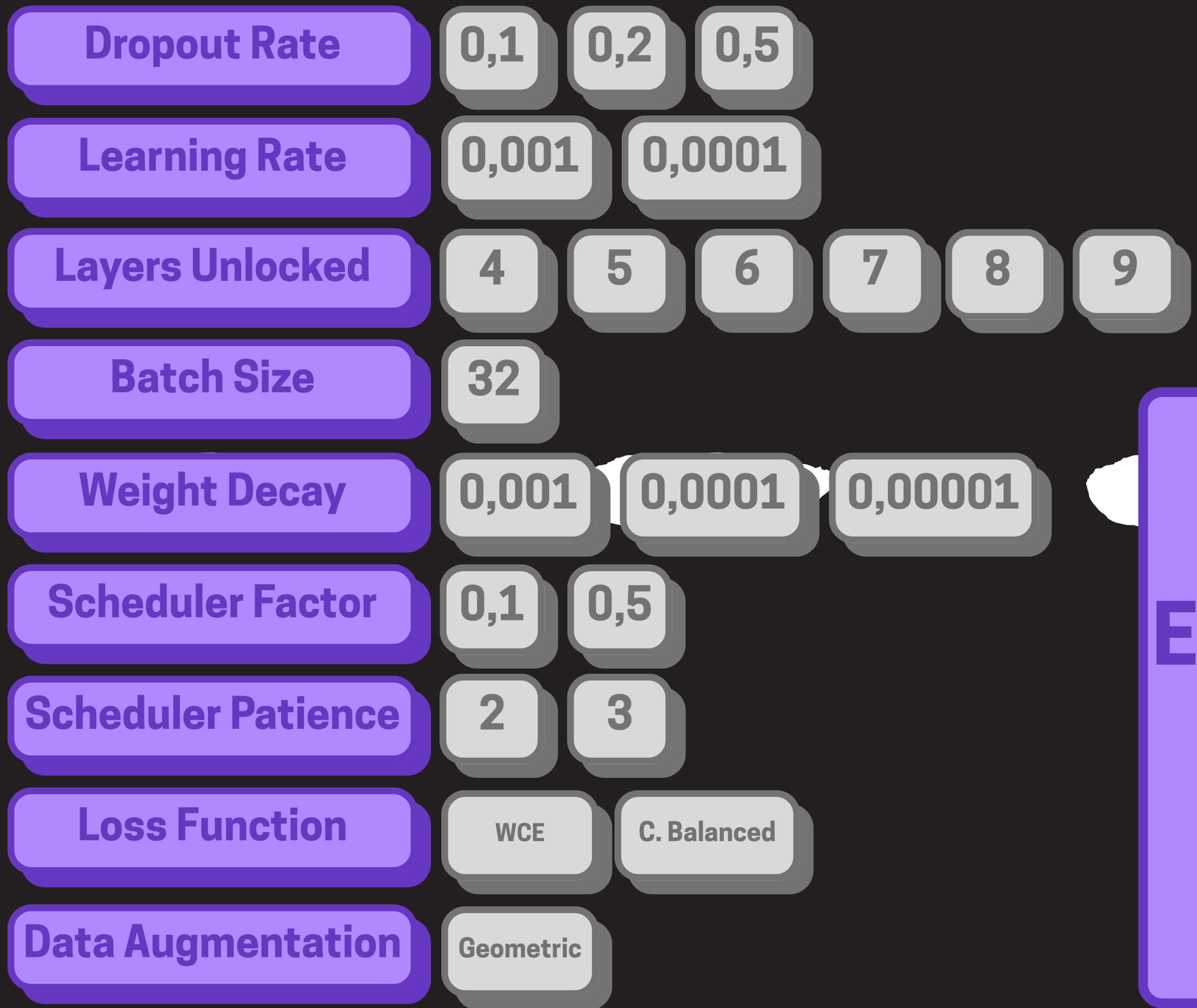
0,0634

Balanced Accuracy

0,9617

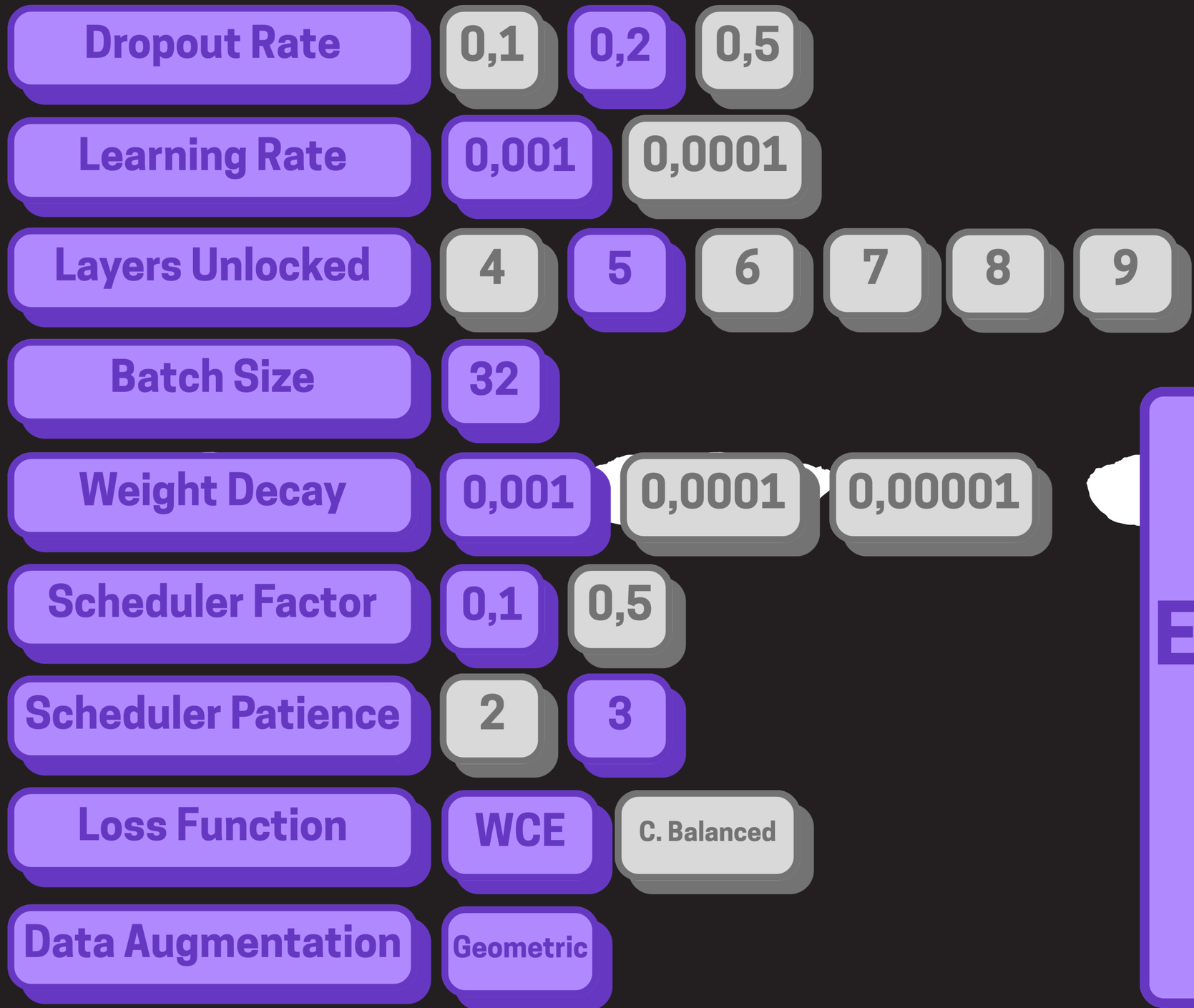
F1-Score

0,9621



EfficientNet-B0

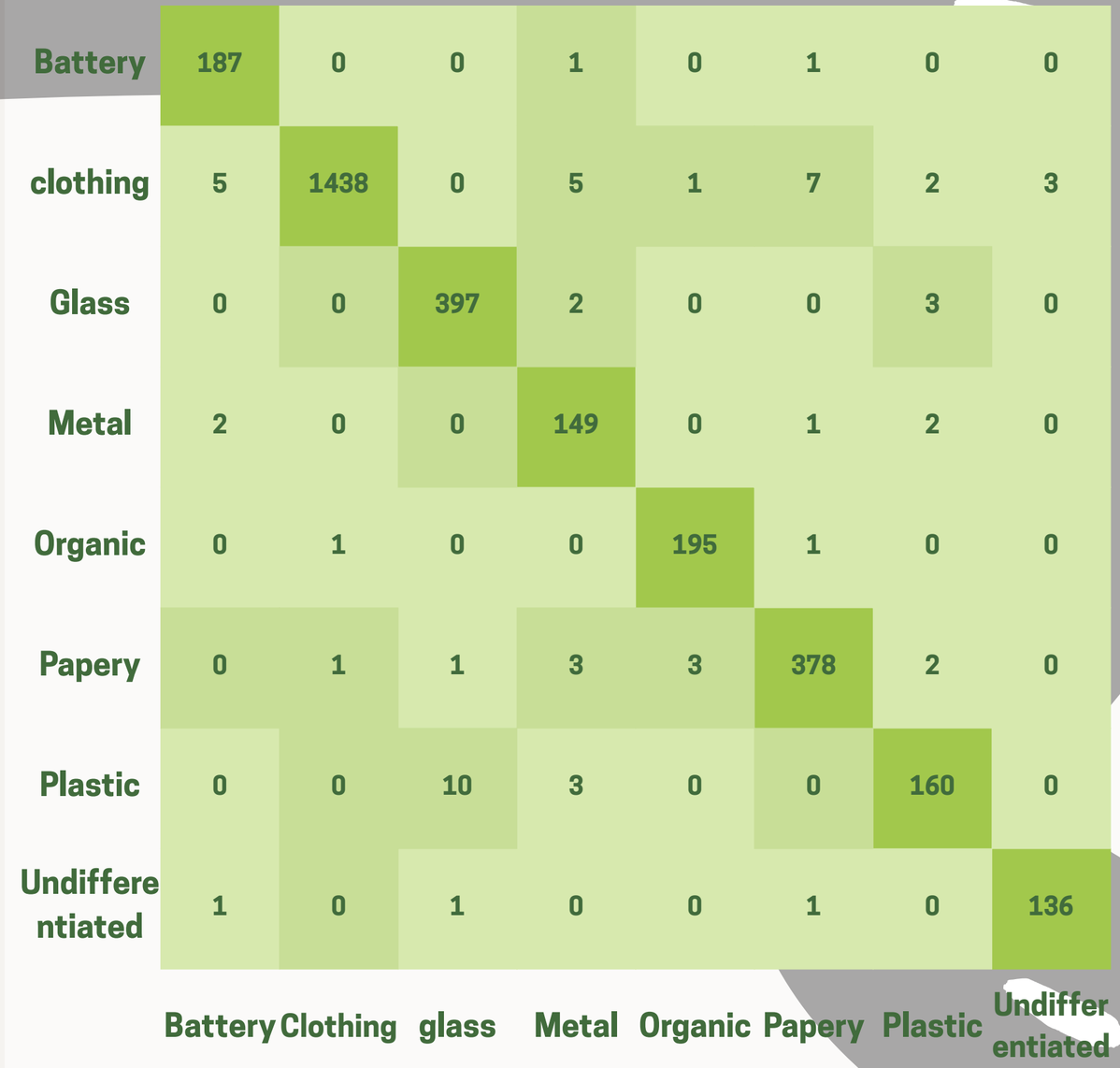
Optimization



EfficientNet-B0

Optimization

OPTIMIZATION ANALYSIS



	Battery	Clothing	glass	Metal	Organic	Papery	Plastic	Undifferentiated
Battery	187	0	0	1	0	1	0	0
clothing	5	1438	0	5	1	7	2	3
Glass	0	0	397	2	0	0	3	0
Metal	2	0	0	149	0	1	2	0
Organic	0	1	0	0	195	1	0	0
Papery	0	1	1	3	3	378	2	0
Plastic	0	0	10	3	0	0	160	0
Undifferentiated	1	0	1	0	0	1	0	136

Validation Loss

0,7712*

Balanced Accuracy

0,9745

F1-Score

0,9695



THANKS FOR ATTENTION







Rebecca Beatrice

Michela Gaeta

Fabio Genito